

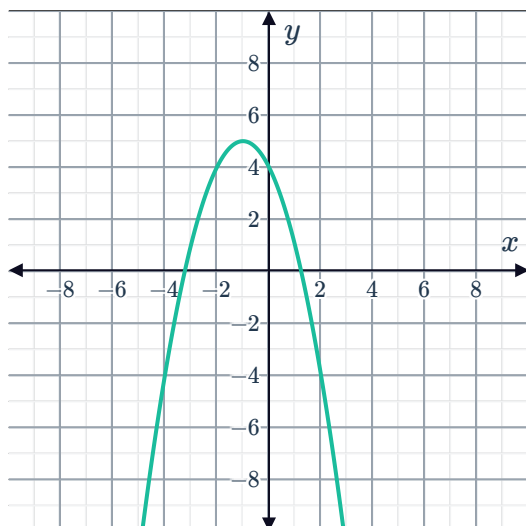
# Quadratic equations and parabolas



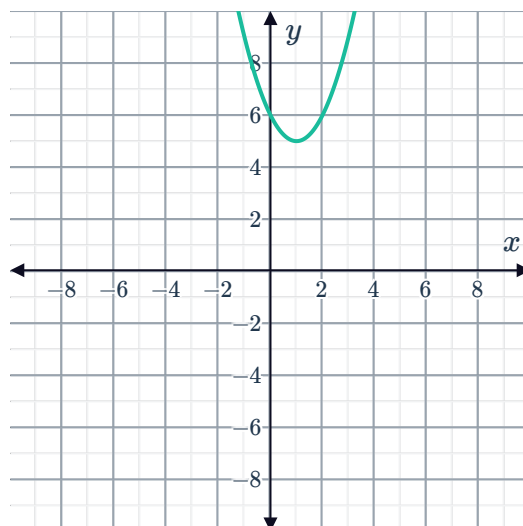
1 For the following graphs of functions of the form  $y = ax^2 + bx + c$  :

- State whether the vertex of the parabola is a maximum or minimum point.
- State whether the value of  $a$  is negative or positive.
- State the number of solutions to the equation  $ax^2 + bx + c = 0$ .

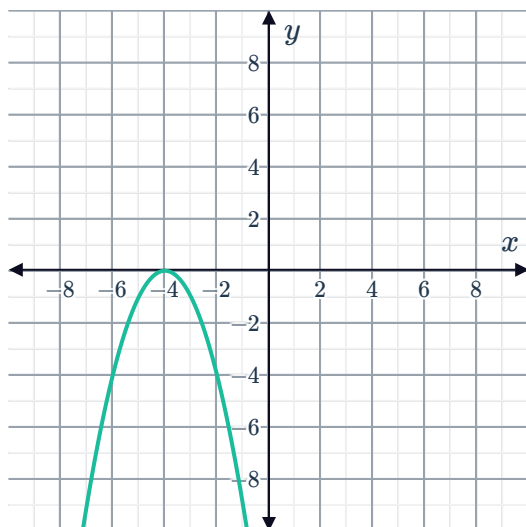
a



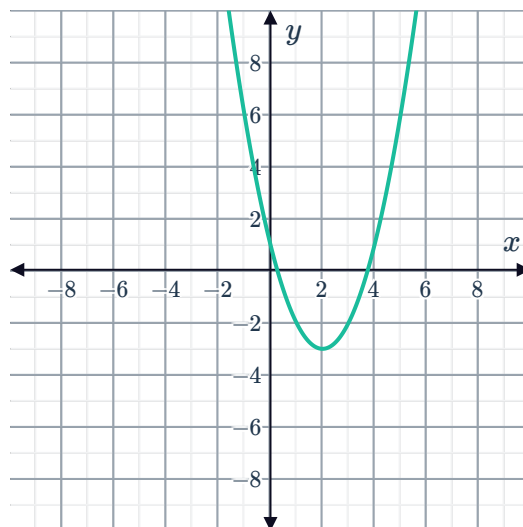
b



c



d



2 Below is the result after using the quadratic formula to solve an equation:

$$x = \frac{-(-10) \pm \sqrt{-1}}{8}$$

What can be concluded about the solutions of the equation?

3 By inspection, determine the number of real solutions for each of the following quadratic equations:

a  $x^2 - 9 = 0$

b  $(x - 4)^2 = 0$

c  $(x - 6)^2 = -2$



## The discriminant and equations

4 For the following quadratic equations:

- i Find the value of the discriminant.
- ii Hence state the number of real solutions for the equation.

a  $x^2 + 6x + 9 = 0$

b  $4x^2 - 6x + 7 = 0$

c  $2x^2 - 2x = x - 1$

d  $x^2 - 4 = 0$

e  $-x^2 - 8x - 16 = 0$

f  $-2x^2 + 5x - 6 = 0$

g  $-x^2 + 6x + 1 = 0$

h  $-x^2 + 25 = 0$

5 Consider the equation  $x^2 + 22x + 121 = 0$ .

- a Find the value of the discriminant.
- b State whether the solutions to the equation are rational or irrational.

6 For the following equations:

- i Find the value of the discriminant.
- ii State the nature of the roots.

a  $5x^2 + 4x + 8 = 0$

b  $4x^2 + 4x - 6 = 0$

c  $4x^2 - 4x + 1 = 0$

d  $x^2 - 16 = 0$

e  $9x^2 - 42x = -49$

f  $2x^2 - 9x = -2$

g  $x^2 = -6x - 10$

h  $\frac{1}{4}x^2 - 2x + \frac{1}{4} = 0$

7 Consider the equation  $x^2 - 8x - 48 = 0$ .

- a Find the discriminant.
- b Describe the nature of the roots.
- c Find the solutions of the equation.

8 For a particular quadratic equation  $b^2 - 4ac = 0$ , what can be said about the solutions of the quadratic equation?

9 The solutions of a quadratic equation are 9 and  $-9$ . What can be said about the value of  $b^2 - 4ac$ ?

10 Find an expression for the discriminant of the following quadratic equations:



a  $mx^2 + 3x - 2 = 0$

b  $x^2 - 4nx - 2 = 0$

c  $x^2 + 5x + p - 5 = 0$

d  $-px^2 + 4x + p = 0$

e  $mx^2 - 2mx + m - 3 = 0$

f  $\frac{3n}{4}x^2 + \frac{1n}{2}x + 2n = 0$

11 Find the range of values of  $a$  for which the quadratic equation  $ax^2 - bx + c = 0$ , where  $b = 6$  and  $c = 10$ , has two distinct real solutions.

12 Consider the equation in terms of  $x$ :

$$mx^2 + 2x - 1 = 0$$

Given that it has two unique solutions, determine the possible values of  $m$ .

13 Consider the equation in terms of  $x$ :

$$mx^2 - 3x - 5 = 0$$

- a Given that the equation has two unique solutions, determine the possible values of  $m$ .
- b State the value of  $m$  that must be eliminated from the range of solutions found in the previous part. Explain your answer.

14 Find the values of  $n$  for which  $x^2 - 8nx + 1296 = 0$  has one solution.

15 Find the value of  $k$  for which  $16x^2 + 8x + k = 0$  has equal roots.

16 Find the values of  $m$  for which  $(m + 4)x^2 + 2mx + 2 = 0$  has a single solution.

17 For the following equations:

- i Find the discriminant in terms of  $k$ .
- ii Find the value of  $k$  so that the equation has equal solutions.
- iii Find the value of  $k$  so that the equation has real solutions.
- iv Find the value of  $k$  so that the equation has no real solutions.
- v Find the value of  $k$  so that the equation has real and distinct solutions.

a  $2x^2 + 8x + k = 0$

b  $kx^2 + 20x + 2 = 0$

c  $(k + 4)x^2 + 10x + 3 = 0$

18 Consider the equation in terms of  $x$ :



$$x^2 + 18x + k + 7 = 0$$

- a Find the values of  $k$  for which the equation has no real solutions.
- b State the smallest possible integer value of  $k$ .

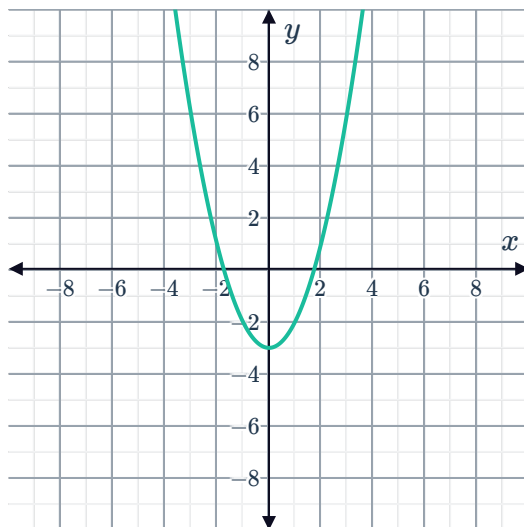
## The discriminant and parabolas

- 19 When graphing a particular parabola, Katrina used the quadratic formula and found that  $b^2 - 4ac = -5$ . How many  $x$ -intercepts does the parabola have?
- 20 When graphing a particular parabola, Tony used the quadratic formula and found that  $b^2 - 4ac = 0$ . How many  $x$ -intercepts does the parabola have?
- 21 For each of the following graphs of a quadratic  $f(x) = ax^2 + bx + c$ , with discriminant  $\Delta = b^2 - 4ac$ :

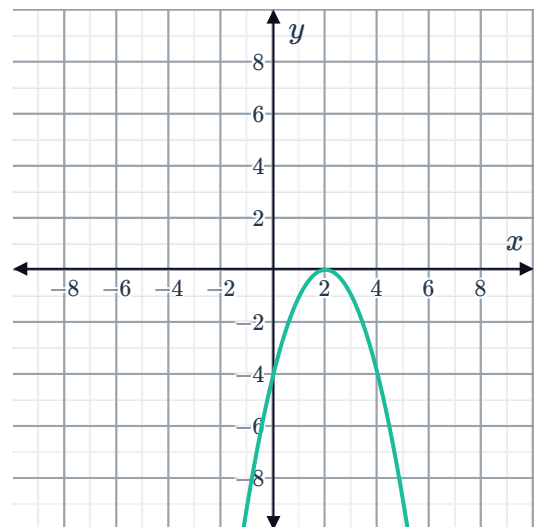
i State whether  $a > 0$  or  $a < 0$ .

ii State whether  $\Delta > 0$ ,  $\Delta < 0$  or  $\Delta = 0$ .

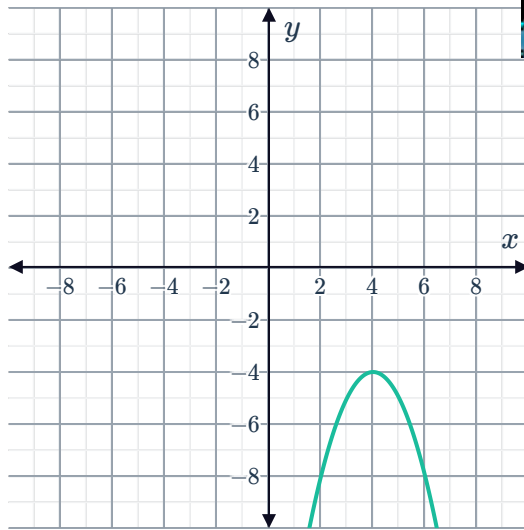
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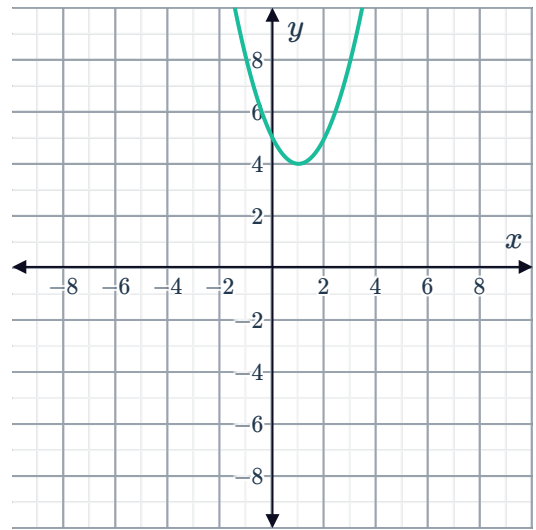
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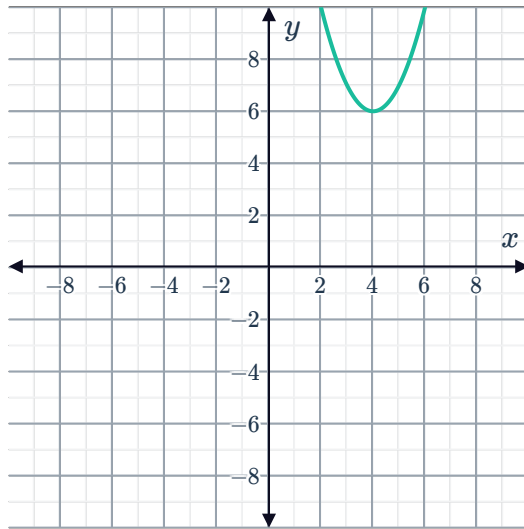
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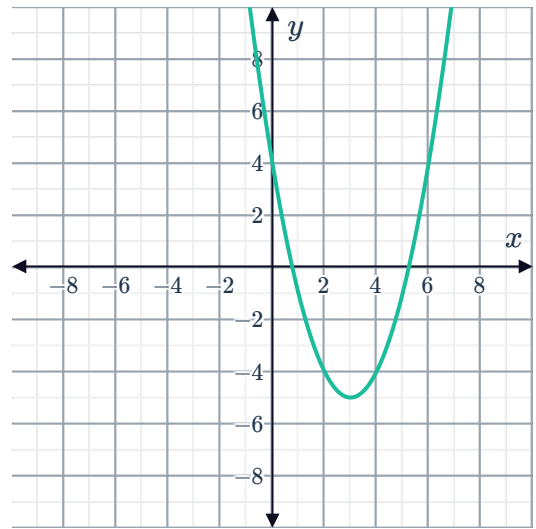
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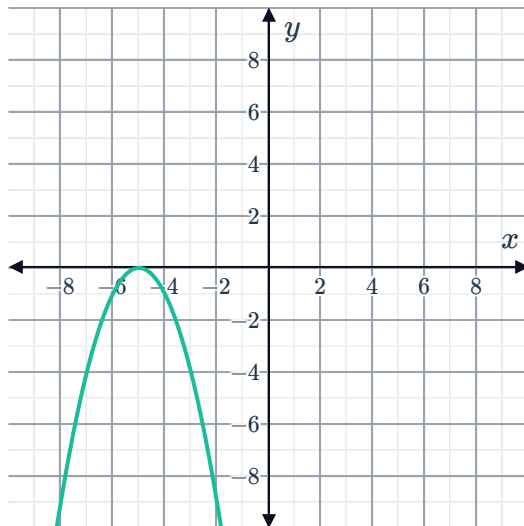
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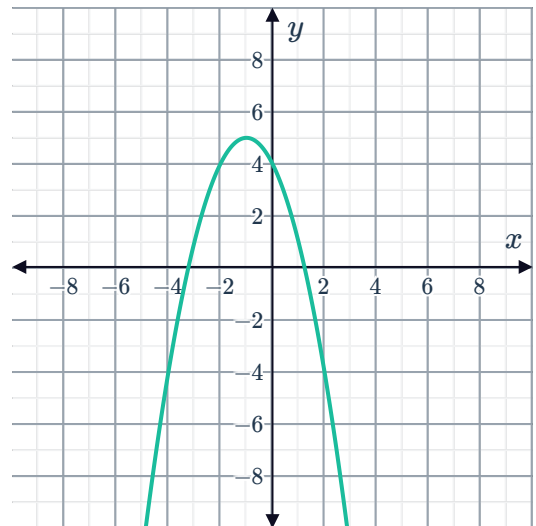
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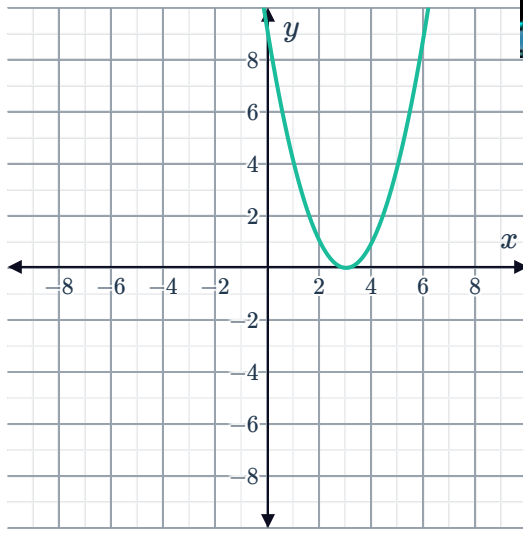
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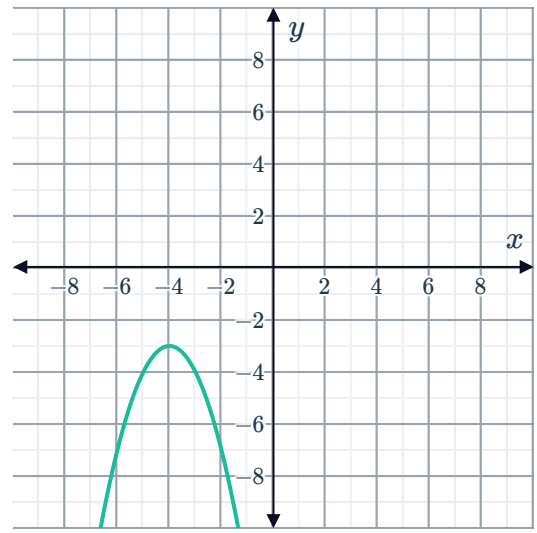
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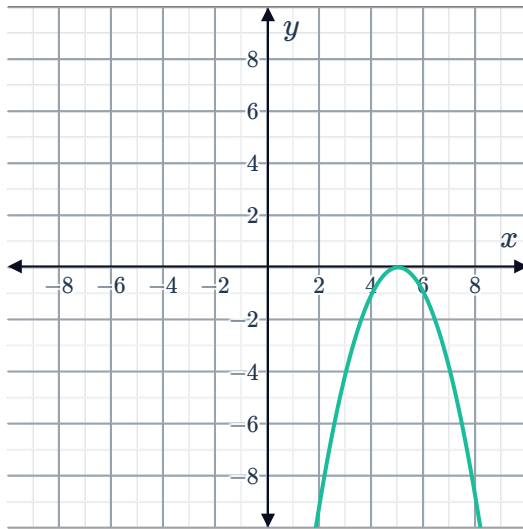
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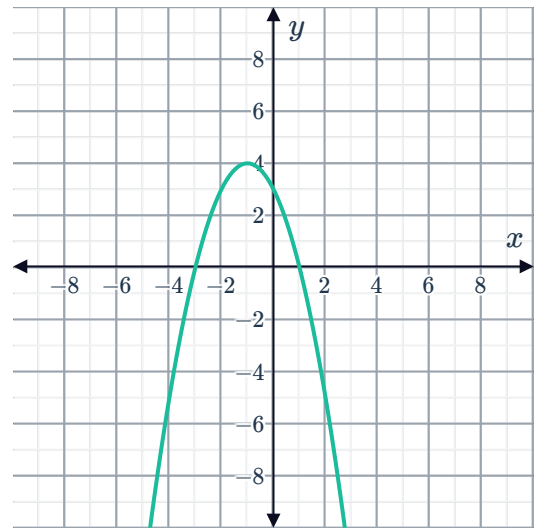
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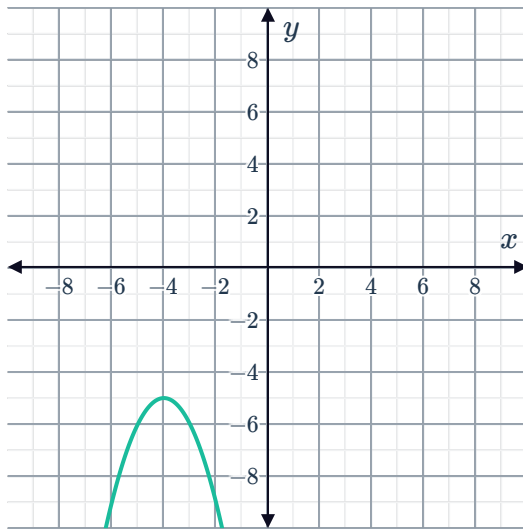
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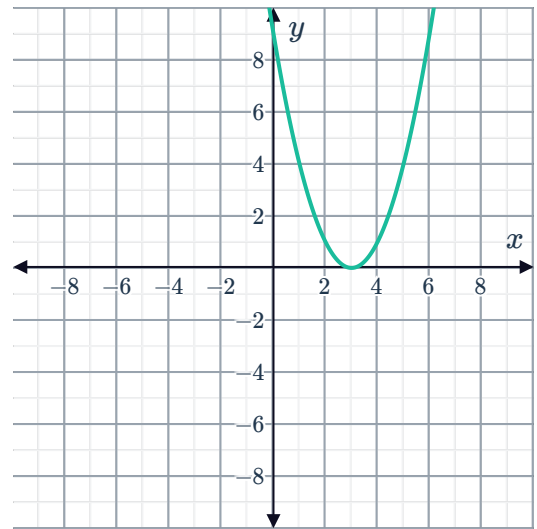
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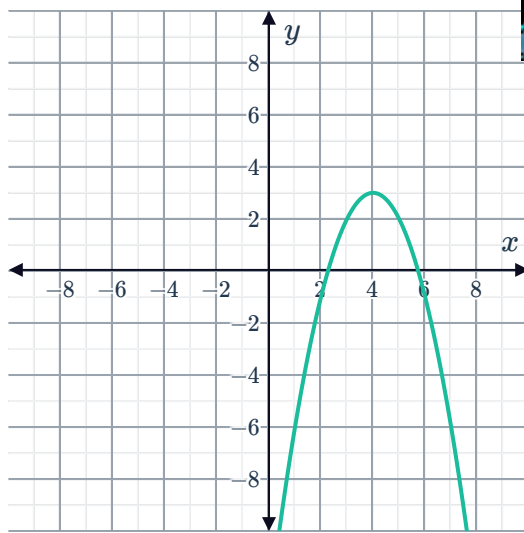
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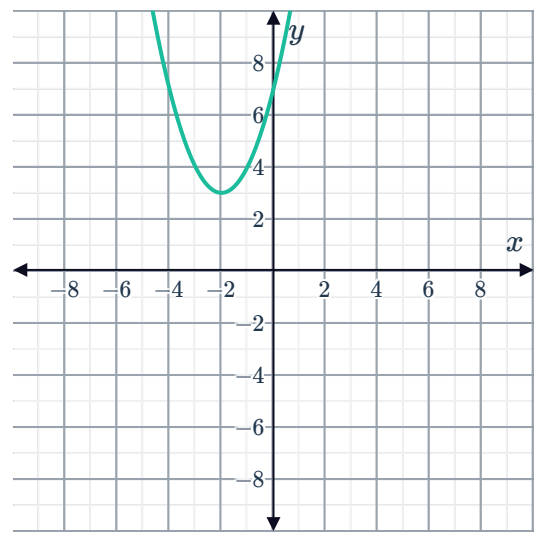
n



o



p



**22** Consider the graph of the quadratic function:

$$y = m - 9x - 3x^2$$

- Find the possible values of  $m$ , if the graph has no  $x$ -intercepts.
- State the largest possible integer value of  $m$ .

**23** Determine the value(s) of  $k$  for which the graph of  $y = 4x^2 - 4x + k - 15$  just touches the  $x$ -axis.

## Application

**24** Consider a right-angled triangle with side lengths  $x$  units,  $x + p$  units and  $x + q$  units, ordered from shortest to longest. No two sides of this triangle have the same length.

- Complete the statement:  
 $p$  and  $q$  have lengths such that  $0 < p < q$ .
- Write a quadratic equation that describes the relationship between the sides of the triangle in terms of  $x$ .
- Find the discriminant of this quadratic equation.
- State the number of real solutions to the quadratic equation. Explain your answer.
- Find the value of  $x$  in terms of  $p$  when  $q = 2p$ .

